Given Data.:

Shutter Width.:12700 mm

Shutter Height.: 2750 mm

Wind Load.: 14km/hr

GI Slat Weight: 650 kg

Intermediate Support tube : 432 kg

A) <u>For Deflection Calculation</u>.:

1.Wind Speed.:

$V = 14{,}000/3600$

$= 3.89$ m/sec.

2.Wind Pressure (P).:

$\mathbf{P} = 0.613 \times V^2$

$= 0.613 \times (3.89)^2$

$= 0.613 \times 15.1321$

Wind Pressure (P) = 9.275 N/m^2 or Pa

3.Convert Given Load.:

Given load = GI Slat Weight + Shaft weight

= 650 + 432

Total Weight = 1082 kg per meter

= 1082/12.7= 81.25 kg per meter

Convert to 'N/m' –

= Mass x Accl^n due to gravity

= 81.25 x 9.81

= 797.15 N/m

Convert to 'N/mm' –

= 797.15 /1000

Total Given Load = 0.797 N/mm

Wind load per meter 'Width'

Area per meter Width = 12.7x2.750

= 34.95m^2

4.Wind Load (Wind force per meter).:

= P x A

= 9.275 x 34.95

= 323.92 N/m

Convert to 'N/mm' -

= 323.92/1000

= 0.323 N/mm

Where:

P = 9.275 N/m^2

A = Width X Height

= 12.7 X 2.750

A = 34.92 m^2

5.Total Weight per 'mm'

Total Weight = Total Given Load

= 0.797 N/mm

6.Section Properties.:

OD = 220 mm

ID = 208 mm

Thickness (t) = 6 mm

d = D-2t

=220-12

d = 208 mm

7.Movement of Inertia (I).:

$I = (\pi (D^4 – d^4))/ 64$

$I = ((3.14*(220^4 - 208^4)) / 64$

$I = 23109668.65\ mm^4$

8.Section Modulus (Z).:

$Z = I / Y$

Where:

$Y = D/2$

$= 220/2 = 110$ mm

$Z = 23109668.65 / 110$

$Z = 210087.89\ mm^3$

9.Deflection Calculation.:

Simply Supported Beam.:

$= 5\ w\ L^4 / 384EI$

$= (5\ x0.797x12700^4) / (384\ x\ 2x10^5\ x\ 23109668.65)$

Deflection =58.41 mm

10.Allowable deflection

For shaft:

Allowable = L/400

= 12900/400

= 32.25 mm

10.Final Result.:

Calculated deflection = 58 mm

Allowable deflection = 32.25mm

The shaft will sag about 58 mm

Actual site shaft deflection =55 mm

B) <u>For Bending Stress Calculation</u>

1.Maximum Bending Movement.:

$M = wL^2/8$

$= (0.797 \times 12700^2)/8$

$= 16068516.25$ N-mm.

2.Section Modulus (Z) = 210087.89 mm^3

3.**Bending Stress ('N/mm').:**

Bending Stress = M/Z

= 16068516.25 / 210087.89

$=76.48$ N/mm^2

=76 M pa.

Allowable Stress = 0.5* fy

Or

Allowable Stress = fy/Factor of Safety

For MS-Yield Strength = 250 MPA & Ultimate Strength = 400 MPA

1) Allowable stress = 0.5 x 400
   = 125 N/mm^2 allowable stress

4.Final Result.:

76 M pa is less than 125 M pa allowable stress

Since**,**

Design is Safe in Bending Stress.